Program Summary - Assignment3.sas

Execution Environment

Author:	u64425566
File:	/home/u64425566/BAN_100/Assignment3.sas
SAS Platform:	Linux LIN X64 5.14.0-284.30.1.el9_2.x86_64
SAS Host:	ODAWS02-USW2-2.ODA.SAS.COM
SAS Version:	9.04.01M8P02222023
SAS Locale:	en_US
Submission Time:	4/17/2026, 1:55:34 PM
Browser Host:	BRAS-BASE-TOROON5049W-GRC-02-142-126-227-211.DSL.BELL.CA
User Agent:	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36 Edg/147.0.0.0
Application Server:	ODAMID00-USW2-2.ODA.SAS.COM

Code: Assignment3.sas

```
/*-------------------------------------------------------
  Load the Fitness dataset
-------------------------------------------------------*/
data fitness;
    input age weight oxy runtime rstpulse runpulse maxpulse;
    case = _n_;
datalines;
44 89.47 44.609 11.37 62 178 182
40 75.07 45.313 10.07 62 185 185
44 85.84 54.297 8.65 45 156 168
42 68.15 59.571 8.17 40 166 172
38 89.02 49.874 9.22 55 178 180
47 77.45 44.811 11.63 58 176 176
40 75.98 45.681 11.95 70 176 180
43 81.19 49.091 10.85 64 162 170
44 81.42 39.442 13.08 63 174 176
38 81.87 60.055 8.63 48 170 186
44 73.03 50.541 10.13 45 168 168
45 87.66 37.388 14.03 56 186 192
45 66.45 44.754 11.12 51 176 176
47 79.15 47.273 10.60 47 162 164
54 83.12 51.855 10.33 50 166 170
49 81.42 49.156 8.95 44 180 185
51 69.63 40.836 10.95 57 168 172
51 77.91 46.672 10.00 48 162 168
48 91.63 46.774 10.25 48 162 164
49 73.37 50.388 10.08 67 168 168
57 73.37 39.407 12.63 58 174 176
54 79.38 46.080 11.17 62 156 165
52 76.32 45.441 9.63 48 164 166
```

```
50 70.87 54.625 8.92 48 146 155
51 67.25 45.118 11.08 48 172 172
54 91.63 39.203 12.88 44 168 172
51 73.71 45.790 10.47 59 186 188
57 59.08 50.545 9.93 49 148 155
49 76.32 48.673 9.40 56 186 188
48 61.24 47.920 11.50 52 170 176
52 82.78 47.467 10.50 53 170 172
;
run;

/*-----------------------------------------------------------
  Scatterplot Matrix (visualizing relationships)
-----------------------------------------------------------*/
proc sgscatter data=fitness;
    matrix age weight oxy runtime rstpulse runpulse maxpulse;
run;

/*-----------------------------------------------------------
  Correlation Matrix + Scatterplot Matrix + Histograms
-----------------------------------------------------------*/
proc corr data=fitness plots=matrix(histogram);
    var age weight oxy runtime rstpulse runpulse maxpulse;
run;

/*-----------------------------------------------------------
  Optional: Cleaner correlation table (Pearson only)
-----------------------------------------------------------*/
proc corr data=fitness pearson;
    var age weight oxy runtime rstpulse runpulse maxpulse;
run;
```

Log: Assignment3.sas

Notes (15)

```
1          OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
68
69         /*-----------------------------------------------------------
70           Load the Fitness dataset
71         -----------------------------------------------------------*/
72         data fitness;
73             input age weight oxy runtime rstpulse runpulse maxpulse;
74             case = _n_;
75         datalines;

NOTE: The data set WORK.FITNESS has 31 observations and 8 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
```

```
      system cpu time     0.00 seconds
      memory              688.62k
      OS Memory           26536.00k
      Timestamp           04/17/2026 05:55:38 PM
      Step Count                        81  Switch Count  2
      Page Faults                       0
      Page Reclaims                     87
      Page Swaps                        0
      Voluntary Context Switches        11
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           264


      ;
      run;

      /*-------------------------------------------------------
        Scatterplot Matrix (visualizing relationships)
      -------------------------------------------------------*/
      proc sgscatter data=fitness;
          matrix age weight oxy runtime rstpulse runpulse maxpulse;
      run;

NOTE: PROCEDURE SGSCATTER used (Total process time):
      real time           0.25 seconds
      user cpu time       0.04 seconds
      system cpu time     0.01 seconds
      memory              8418.03k
      OS Memory           32680.00k
      Timestamp           04/17/2026 05:55:38 PM
      Step Count                        82  Switch Count  1
      Page Faults                       0
      Page Reclaims                     1865
      Page Swaps                        0
      Voluntary Context Switches        207
      Involuntary Context Switches      2
      Block Input Operations            0
      Block Output Operations           624

NOTE: Some of the tick values have been thinned.
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NOTE: There were 31 observations read from the data set WORK.FITNESS.

      /*-------------------------------------------------------
        Correlation Matrix + Scatterplot Matrix + Histograms
      -------------------------------------------------------*/
      proc corr data=fitness plots=matrix(histogram);
          var age weight oxy runtime rstpulse runpulse maxpulse;
```
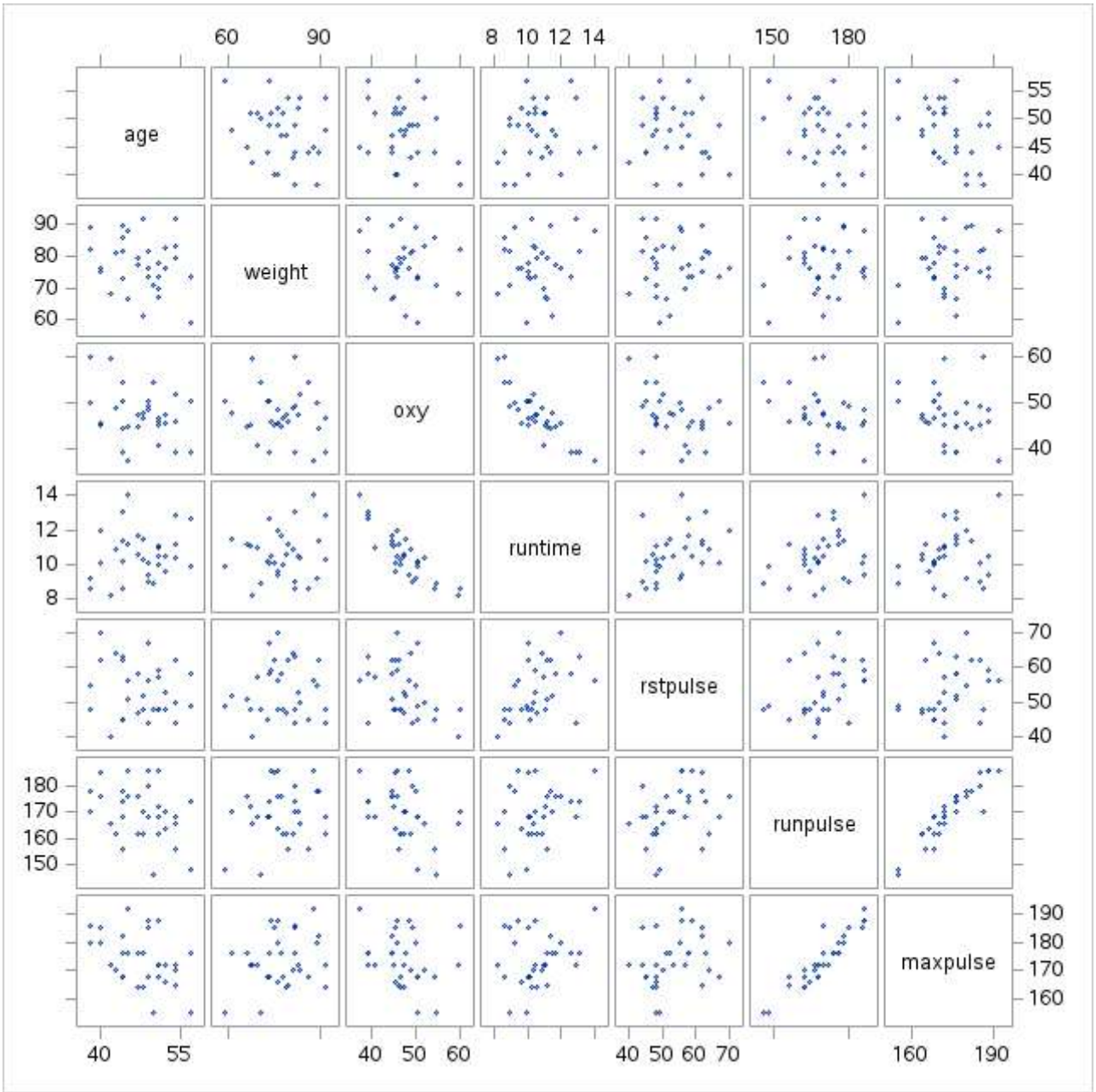

```
122         run;

NOTE: PROCEDURE CORR used (Total process time):
      real time           0.17 seconds
      user cpu time       0.08 seconds
      system cpu time     0.00 seconds
      memory              4022.71k
      OS Memory           34140.00k
      Timestamp           04/17/2026 05:55:38 PM
      Step Count                        83  Switch Count  0
      Page Faults                       0
      Page Reclaims                     527
      Page Swaps                        0
      Voluntary Context Switches        313
      Involuntary Context Switches      3
      Block Input Operations            0
      Block Output Operations           224


123
124         /*-------------------------------------------------------
125           Optional: Cleaner correlation table (Pearson only)
126         -------------------------------------------------------*/
127         proc corr data=fitness pearson;
128             var age weight oxy runtime rstpulse runpulse maxpulse;
129         run;

NOTE: PROCEDURE CORR used (Total process time):
      real time           0.05 seconds
      user cpu time       0.05 seconds
      system cpu time     0.01 seconds
      memory              833.46k
      OS Memory           32680.00k
      Timestamp           04/17/2026 05:55:38 PM
      Step Count                        84  Switch Count  0
      Page Faults                       0
      Page Reclaims                     55
      Page Swaps                        0
      Voluntary Context Switches        3
      Involuntary Context Switches      2
      Block Input Operations            0
      Block Output Operations           16


130
131
132         OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
142
```

Results: Assignment3.sas

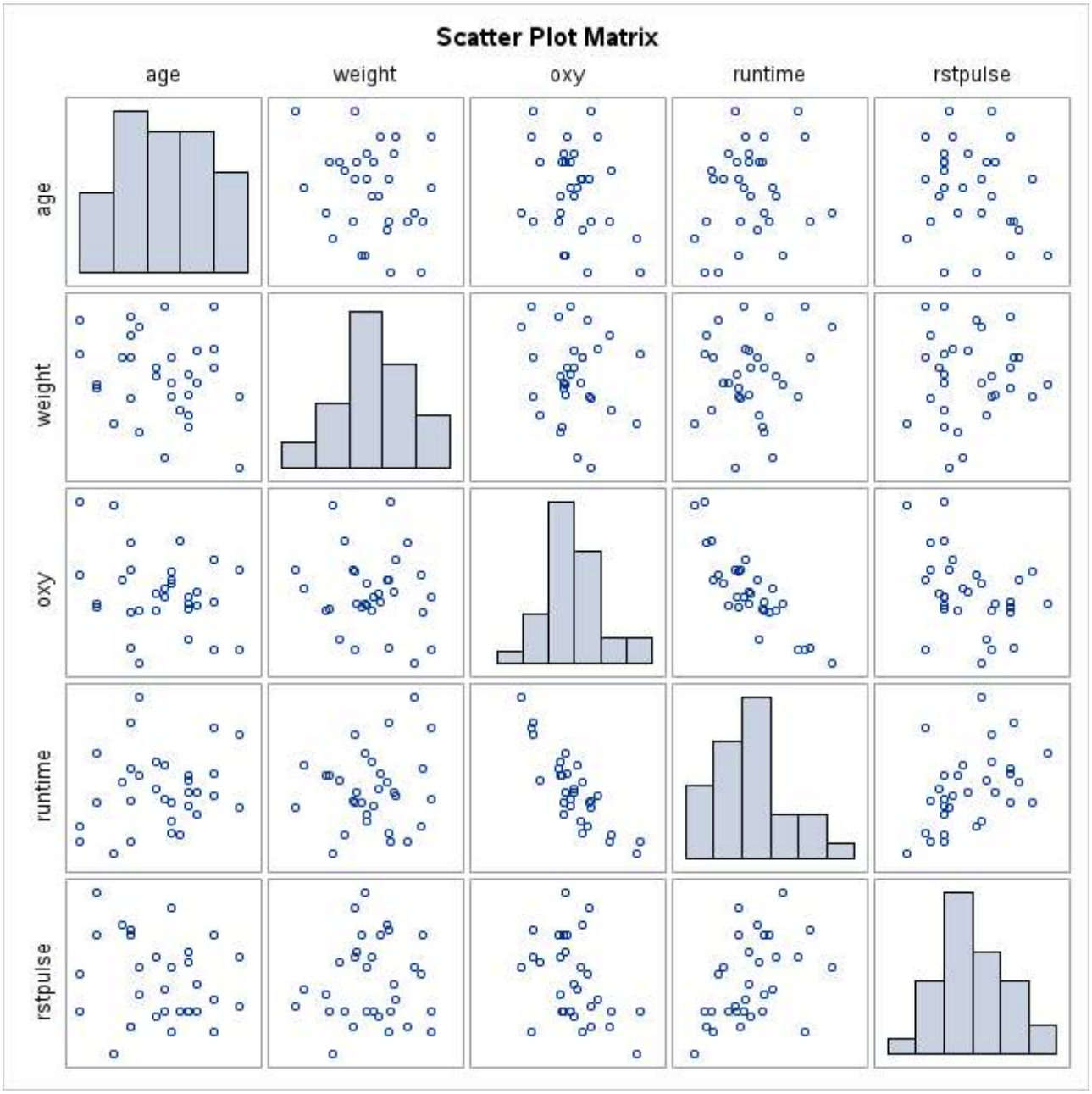


The CORR Procedure

7 Variables:	age weight oxy runtime rstpulse runpulse maxpulse

Simple Statistics						
Variable	**N**	**Mean**	**Std Dev**	**Sum**	**Minimum**	**Maximum**
age	31	47.67742	5.21144	1478	38.00000	57.00000
weight	31	77.44452	8.32857	2401	59.08000	91.63000
oxy	31	47.37581	5.32723	1469	37.38800	60.05500
runtime	31	10.58613	1.38741	328.17000	8.17000	14.03000
rstpulse	31	53.45161	7.61944	1657	40.00000	70.00000
runpulse	31	169.64516	10.25199	5259	146.00000	186.00000
maxpulse	31	173.77419	9.16410	5387	155.00000	192.00000

Pearson Correlation Coefficients, N = 31 Prob > |r| under H0: Rho=0							
	age	**weight**	**oxy**	**runtime**	**rstpulse**	**runpulse**	**maxpulse**
age	1.00000	-0.23354 0.2061	-0.30459 0.0957	0.18875 0.3092	-0.16410 0.3777	-0.33787 0.0630	-0.43292 0.0150
weight	-0.23354 0.2061	1.00000	-0.16275 0.3817	0.14351 0.4412	0.04397 0.8143	0.18152 0.3284	0.24938 0.1761
oxy	-0.30459 0.0957	-0.16275 0.3817	1.00000	-0.86219 <.0001	-0.39936 0.0260	-0.39797 0.0266	-0.23674 0.1997
runtime	0.18875 0.3092	0.14351 0.4412	-0.86219 <.0001	1.00000	0.45038 0.0110	0.31365 0.0858	0.22610 0.2213
rstpulse	-0.16410 0.3777	0.04397 0.8143	-0.39936 0.0260	0.45038 0.0110	1.00000	0.35246 0.0518	0.30512 0.0951
runpulse	-0.33787 0.0630	0.18152 0.3284	-0.39797 0.0266	0.31365 0.0858	0.35246 0.0518	1.00000	0.92975 <.0001
maxpulse	-0.43292 0.0150	0.24938 0.1761	-0.23674 0.1997	0.22610 0.2213	0.30512 0.0951	0.92975 <.0001	1.00000



The CORR Procedure

7 Variables:	age weight oxy runtime rstpulse runpulse maxpulse

Simple Statistics						
Variable	N	Mean	Std Dev	Sum	Minimum	Maximum
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Pearson Correlation Coefficients, N = 31 Prob > |r| under H0: Rho=0							
	age	weight	oxy	runtime	rstpulse	runpulse	maxpulse
age	1.00000	-0.23354 0.2061	-0.30459 0.0957	0.18875 0.3092	-0.16410 0.3777	-0.33787 0.0630	-0.43292 0.0150
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oxy	-0.30459 0.0957	-0.16275 0.3817	1.00000	-0.86219 <.0001	-0.39936 0.0260	-0.39797 0.0266	-0.23674 0.1997
runtime	0.18875 0.3092	0.14351 0.4412	-0.86219 <.0001	1.00000	0.45038 0.0110	0.31365 0.0858	0.22610 0.2213
rstpulse	-0.16410 0.3777	0.04397 0.8143	-0.39936 0.0260	0.45038 0.0110	1.00000	0.35246 0.0518	0.30512 0.0951
runpulse	-0.33787 0.0630	0.18152 0.3284	-0.39797 0.0266	0.31365 0.0858	0.35246 0.0518	1.00000	0.92975 <.0001
maxpulse	-0.43292 0.0150	0.24938 0.1761	-0.23674 0.1997	0.22610 0.2213	0.30512 0.0951	0.92975 <.0001	1.00000